



MINING DEALER BEST PRACTICE

Adjusting the LED Lamps Intensity on the 794 Truck's Dash Panel for Nighttime Operation

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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July 2023
1.11-230726-1

1.0 Introduction

The back lighting intensity of the dashboard signal lamps does not have another level. Signaling LEDs have a strong light for nighttime operation, and some operators complain of eye discomfort due to the brightness. As a result, they paint or place stickers on the panel to reduce the light. The addition of a potentiometer will allow the operator to manually control the LED lighting intensity.



Illustration 1: Signal lamps cap painted by operators.

2.0 Best Practice Description

By installing the potentiometer in the LED's electrical circuit, it was possible to adjust the lamp's intensity. We installed the improvement in a truck and monitored the results and the operator's opinion. It has improved the comfort of operation at nighttime and as result of reducing the brightness there have been no more complaints from operators.



Illustration 2: Intensity levels.

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3.0 Implementation Steps

We elaborated the step-by-step to implement the improvement. The first step was searching for parts compatible with the equipment's electrical system and specifications. The parts defined were potentiometer #610-2112 and knob #432-2913. Both parts already are used in the equipment system.



Illustration 3: Parts

We design the electrical schematic to support our technicians in this job. We installed the potentiometer in series with the negative wire 200-Y1 BK-20 cable. We used the pin A and B.

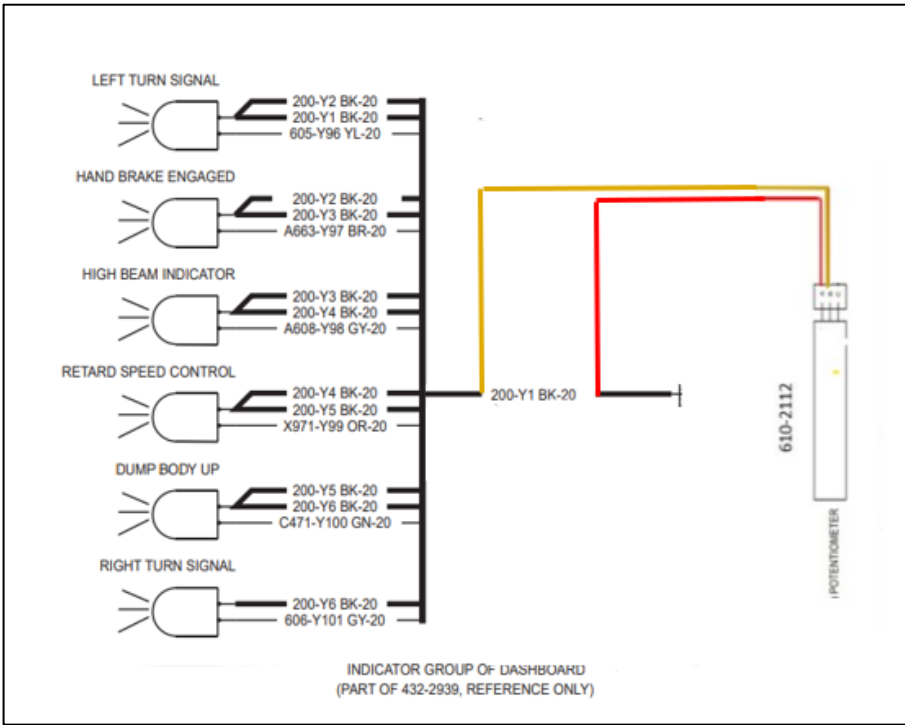


Illustration 4: Schematic

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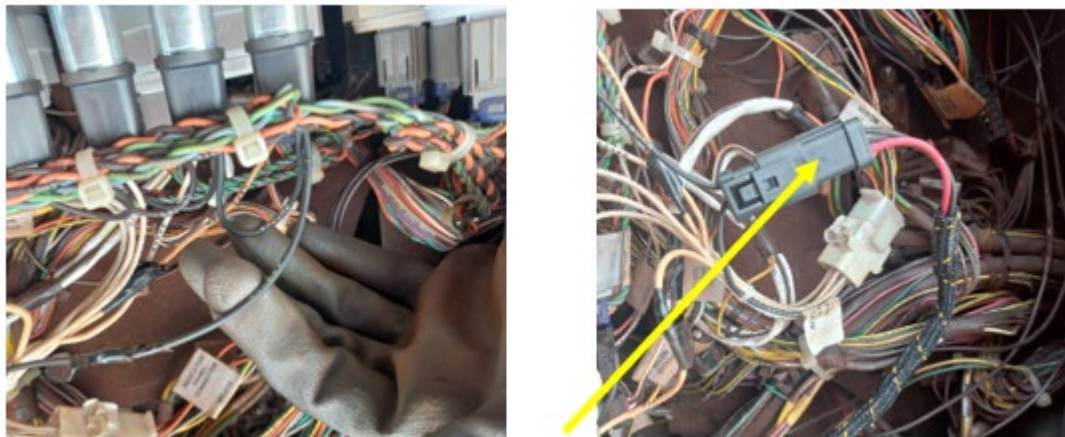


Illustration 5: Electrical installation

The potentiometer and knob were installed on the dash panel, see the image below.

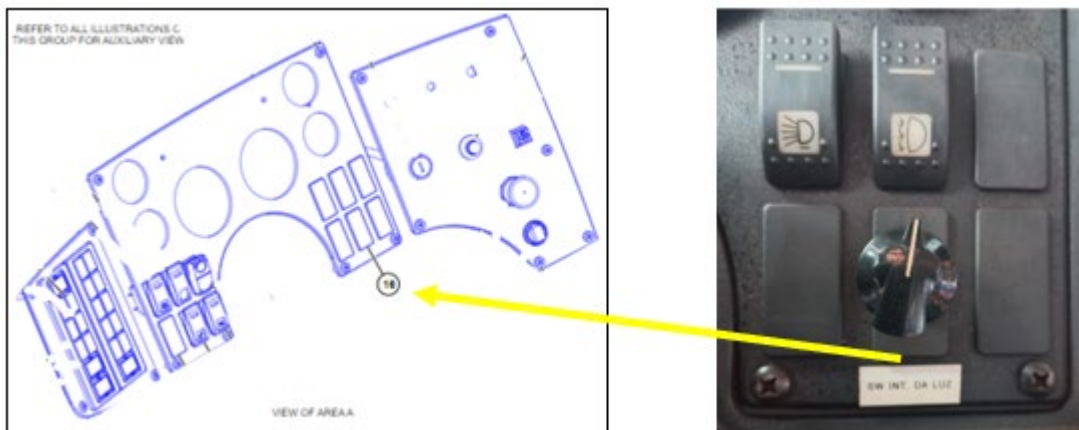


Illustration 6: Potentiometer and Knob

4.0 Benefits

- Reduce ergonomic discomfort for the operator's eyes during the night.
- Quick regulation of the intensity levels of the LED's
- Avoid damage to the panel with paints and adhesives to reduce brightness.
- Low cost

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5.0 Resources Required

PART NUMBER	DESCRIPTION.	QUANTITY.
610-2112	Potentiometer	1
432-2913	Knob	1

Table 1: Parts

6.0 Supporting Attachments / References

Instrument Panel - Remove and Install (UENR2271-23)

7.0 Related Best Practices

NA.

8.0 Acknowledgements

We would like to thank all the professionals involved in this development. A simple, low-cost idea with positive results. Congratulations to everyone involved in the development of the solution.

SOTREQ would also like to thank the indispensable technical contribution of **Doug Friedrich** of **Caterpillar Inc.**

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